AIVOA PHARMACEUTICALS MANUFACTURING CORP.
GLOBAL QUALITY OPERATIONS & COMPLIANCE DIVISION

PHARMACEUTICAL API QUALITY DEVIATION & INVESTIGATION REPORT

Deviation Number:	DEV-2026-1299	**Classification:**	Critical / Particle Size Distribution (PSD)
Product / API Name:	Apixaban Micronized API (USP)	**Batch / Lot Number:**	APX-2026-512
Manufacturing Stage:	Stage 5: Fluid Energy Jet Milling & Packaging	**Equipment Tag:**	Spiral Jet Mill JM-02 / Malvern Mastersizer 3000
Date of Occurrence:	2026-10-15 15:30 UTC	**Date of Discovery:**	2026-10-16 11:25 UTC
Material / Lot ID:	Unmicronized Feed Cake LOT-APX-U512	**Reporting Unit:**	Particle Characterization Unit

1. OBSERVED DEVIATION & ANALYTICAL TEST RESULTS

Test Parameter	Regulatory Specification	Observed Lab Result	Test Methodology / Instrument
Particle Size: d90	**d90 NMT 25.0 µm**	**42.6 µm [OOS]**	Laser Diffraction (Malvern Mastersizer)
Particle Size: d50	**d50 NMT 10.0 µm**	**18.4 µm [OOS]**	Laser Diffraction Dry Dispersion
Particle Size: d10	d10 NMT 3.0 µm	2.8 µm	Laser Diffraction
Assay (Anhydrous)	98.0% - 102.0%	100.2%	HPLC UV 254nm

2. INCIDENT CHRONOLOGY & FORENSIC NARRATIVE

Laser diffraction particle sizing on micronized Apixaban batch APX-2026-512 revealed coarse particle distribution with d90 of 42.6 µm versus specification <= 25.0 µm. Jet mill JM-02 pneumatic telemetry demonstrated that the primary high-pressure nitrogen grinding regulator experienced a diaphragm rupture at 15:30 UTC, causing nitrogen grinding pressure to drop from 7.5 bar to 4.8 bar while feed rate remained continuous at 12 kg/hr.

3. IMMEDIATE CONTAINMENT ACTIONS TAKEN

1. 350 kg of micronized Apixaban in 14 fiber drums placed under QA hold.
2. Jet mill JM-02 locked out; replacement of pressure regulator REG-02 completed.
3. Automated low-pressure grinding cut-off interlock recommended for engineering design.

Prepared by:	T. Larsson (Particle Physicist)	2026-10-16
Verified by:	M. Vance (Production)	2026-10-16
QA Approval:	Dr. E. Gomez (QA Lead)	**Status: PENDING ROOT CAUSE ANALYSIS**